AI QUANTITY SURVEYOR · HACKATHON DEMO

QS Cost — System of Record

See cost trouble **one reporting period early** — while the money is still unspent.

GREEN → AMBER   we call the tip *before* it happens

ASP.NET Core 8 | React + Vite | Postgres · RLS | Claude copilot | **Tower X** · 173 cost centres · AED

TEAM

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THE PROBLEM

A QS spots the overrun weeks too late.

Working in spreadsheets, the problem surfaces at month-end close — **after the invoices are paid and the money is gone.**

- Every cost centre has a planned budget vs earned value vs actual spend.
- Drift hides in the noise until a period closes.
- By then it's a report, not a decision.

THE CONCRETE POUR

~~50,000~~ → 65,000

A concrete pour budgeted at 50,000 AED that quietly became 65,000 should have been flagged **the moment it started drifting** — not at month-end.

OUR INSIGHT

Flag it one period before it tips.

We don't wait for a cost centre to go AMBER. We rank the still-GREEN centres **most likely to tip next period** — while the budget is unspent and the QS can still act.

GREEN at period p	→	AMBER at p + 1
on budget, today	we predict the tip	CPI breaches 0.95

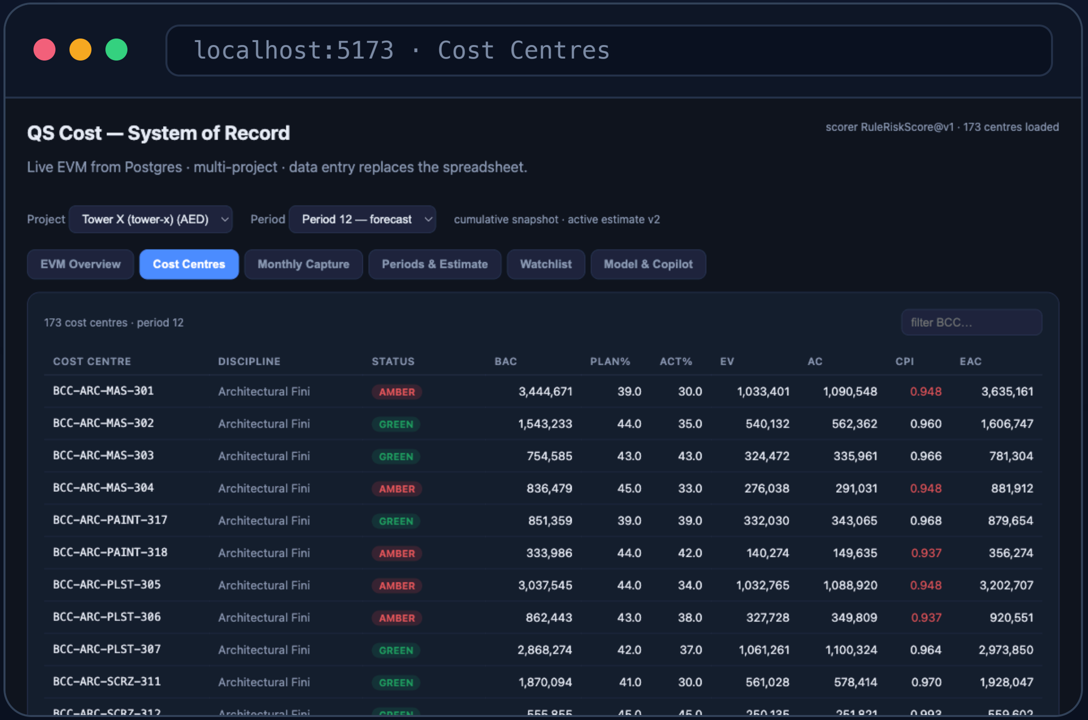
One honest, verifiable event — not a vague "risk." On the Tower X history, **AMBER ≡ CPI < 0.95** on every live row (0 mismatch): the target is a real threshold crossing, so the prediction is gradeable.

THE PRODUCT

The spreadsheet, replaced.

A live **system of record** for EVM — computed from Postgres, multi-project, always current.

- **173 cost centres** with live BAC · Plan% · Actual% · EV · AC · **CPI** · EAC.
- GREEN / AMBER status the moment a number changes.
- Data entry writes to Postgres and EVM recomputes — no month-end batch.



COST CENTRE	DISCIPLINE	STATUS	BAC	PLAN%	ACT%	EV	AC	CPI	EAC
BCC-ARC-MAS-301	Architectural Fini	AMBER	3,444,671	39.0	30.0	1,033,401	1,090,548	0.948	3,635,161
BCC-ARC-MAS-302	Architectural Fini	GREEN	1,543,233	44.0	35.0	540,132	562,362	0.960	1,606,747
BCC-ARC-MAS-303	Architectural Fini	GREEN	754,585	43.0	43.0	324,472	335,961	0.966	781,304
BCC-ARC-MAS-304	Architectural Fini	AMBER	836,479	45.0	33.0	276,038	291,031	0.948	881,912
BCC-ARC-PAINT-317	Architectural Fini	GREEN	851,359	39.0	39.0	332,030	343,065	0.968	879,654
BCC-ARC-PAINT-318	Architectural Fini	AMBER	333,986	44.0	42.0	140,274	149,635	0.937	356,274
BCC-ARC-PLST-305	Architectural Fini	AMBER	3,037,545	44.0	34.0	1,032,765	1,088,920	0.948	3,202,707
BCC-ARC-PLST-306	Architectural Fini	AMBER	862,443	43.0	38.0	327,728	349,809	0.937	920,551
BCC-ARC-PLST-307	Architectural Fini	GREEN	2,868,274	42.0	37.0	1,061,261	1,100,324	0.964	2,973,850
BCC-ARC-SCRZ-311	Architectural Fini	GREEN	1,870,094	41.0	30.0	561,028	578,414	0.970	1,928,047

1 EARLY-WARNING WATCHLIST

A ranked triage list.

> *"Which GREEN cost centres are about to tip to AMBER?"*

- Scored by a **frozen, transparent rule** — `RuleRiskScore@v1`, no black box.
- `0.7 · spend-vs-progress gap + 0.3 · CPI-proximity`
- Click any row → the **cost-centre drawer** opens: variance + AI correction actions.

TOP GREEN CENTRES AT RISK · PERIOD 12

CENTRE	DISCIPLINE	CPI	RISK
BCC-IND-QC-1805	Indirect/Prelims	0.950	+3.29
BCC-IND-SAF-1804	Indirect/Prelims	0.951	+3.22
BCC-ARC-PLST-305	Architectural	0.952	+2.87
BCC-ARC-MAS-302	Architectural	0.960	+2.41
BCC-STR-RC-118	Structural	0.958	+2.18

Risk = spending points ahead of progress, weighted toward centres already near the 0.95 line.

THE PROOF · MONEY SHOT

We let the app grade itself.

The Proof tab rewinds to an earlier period, shows the model **only the past**, lets it flag the top-5 GREEN centres — then reveals what actually happened next month.

Rewind

to period 5 — the model sees only history up to here, no leakage.

Flag

the 5 GREEN centres most likely to tip AMBER next period.

Reveal

step forward one month → the app scores its own calls, hit by hit.

> "Caught a full reporting period early — **before the invoices landed.**"

VALIDATION · HONESTY

Validated, not asserted.

OUR RULE

45%

precision@5 — of the 5 we told the QS to chase, how many really tipped

vs

BEST CPI BASELINE

35%

the strongest CPI-native heuristic we could build

8 rolling-origin folds

117 real GREEN→AMBER transitions

74 centres · 12 periods

Leakage-safe — each period scored by a model trained only on its past

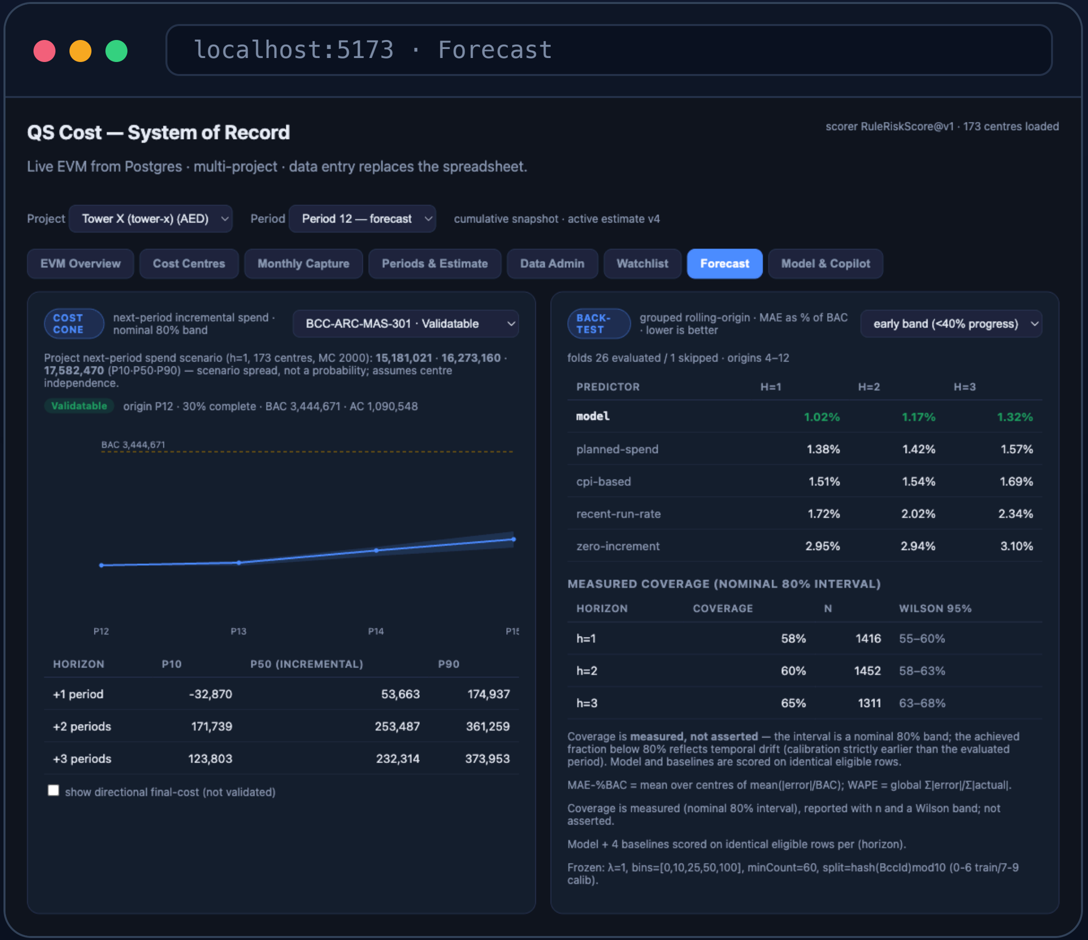
The rule is **frozen and predeclared**, so nothing is tuned on the eval folds — the back-test is unbiased. And we **rejected the naive final-cost EAC**: this project has almost no finished work, so a final-cost number couldn't be honestly validated. Single project — Tower X's number, not a universal benchmark.

2 COST-TRAJECTORY FORECASTER

Next-period spend, with a real band.

> *"How much will we spend next period — and what's the range?"*

- Ridge P50 forecast + a **split-conformal P10–P90 band**.
- Coverage is **measured, not assumed** — reported with the achieved fraction.
- Beats **4 baselines** (planned-spend, CPI-based, run-rate, zero) on MAE-%BAC.

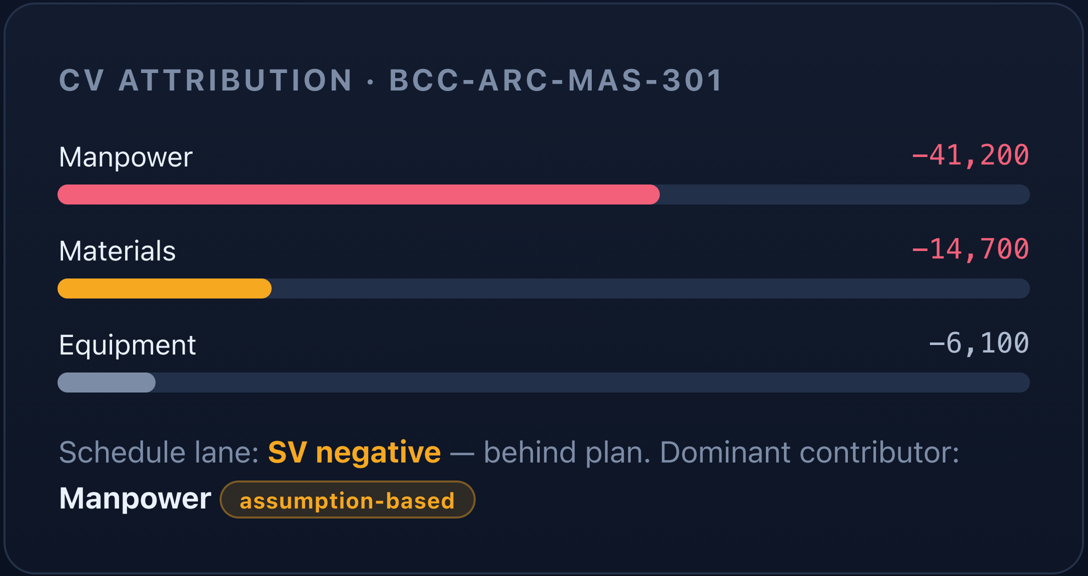


3 VARIANCE ATTRIBUTION BRIDGE

Where is the over-run coming from?

> *"Which resource is this package's variance attributed to?"*

- Decomposes CV into **resource-category lanes** + a schedule (SV) lane.
- Ties out additively: $\Sigma\ \mathtt{CV}_r + \mathtt{residual} == \mathtt{CV}$.
- The *cause* is labelled a **hypothesis** — the data can't split price from productivity.



4 ESTIMATE ASSUMPTION STRESS TEST

Catch trouble **before award.**

The angle others miss — attack cost risk **before a single dirham is spent**, by reviewing the estimate's own assumptions.

CLASS 1

Reconciliation tie-out

Does the resource-level estimate roll up to the BOQ contract total? A hard cross-check.

CLASS 2

Assumption flags

Cohort-gated: aggressive Output Norm, thin unit rate, thin contingency vs peers.

CLASS 3

Peer benchmark

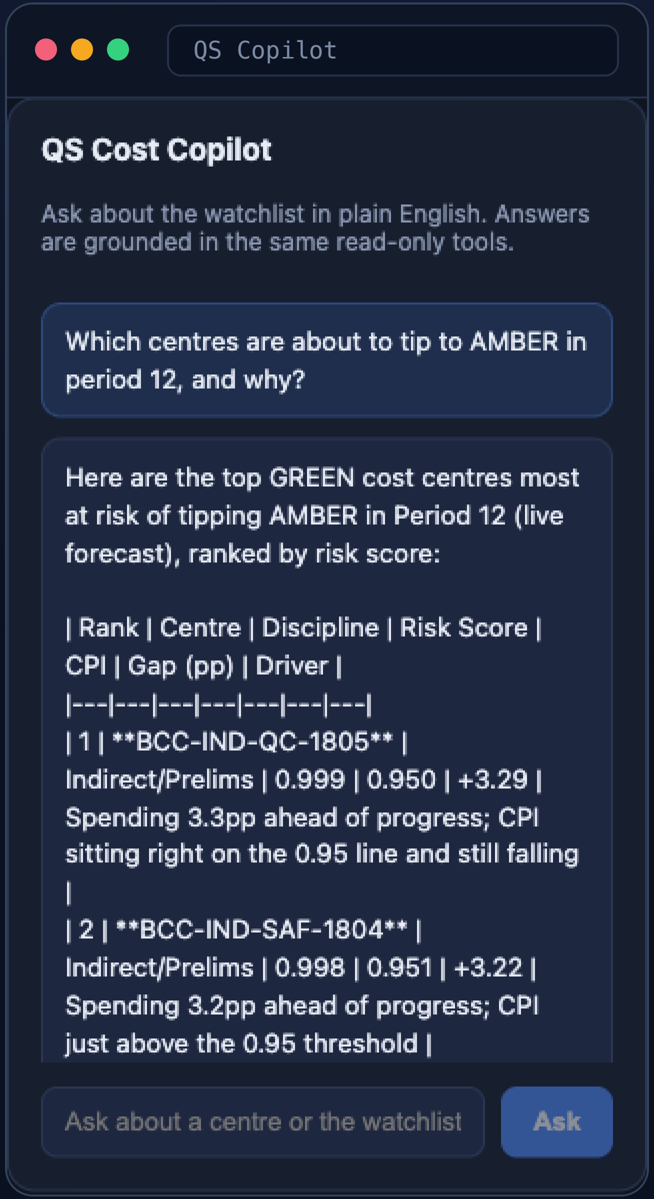
Retrospective only — where this package sits against comparable ones. Honest scope.

Not an "objective under-pricing" oracle — one project can't prove that. It surfaces assumptions **for a QS to review**, with the evidence attached.

Ask it in plain English.

"Tools compute, the model narrates."

- **12 read-only tools** do every calculation — the model never does the AED math. One of them is the **unit-rate what-if**. →
- On an AMBER centre it **drafts prioritised correction actions** + a 3-month **what-if savings band** — the AI supplies only the reduction %, the engine does the AED.
- Ask why **any** centre drifted — an already-AMBER one returns its **trajectory**: when it crossed 0.95, how long it's held, which way CPI moves.
- A **Sources panel** cites every tool call — resolved period/filter, rows excluded, source row IDs.
- Claude Sonnet 5 · Microsoft Agent Framework; **auth-before-LLM**, refuses off-topic — no hallucinated money.



6 UNIT-RATE WHAT-IF

"What if we renegotiate?"

> *"If we get the subcontractor to 299 AED/unit from next period — where does this centre land?"*

IN The assumption

The QS's rate — **not a figure from the sheets**. Echoed back with its effective period, so the assumption is auditable.

ENGINE Reprice remaining

Remaining quantity repriced at that rate, **holding recent physical pace** — a price renegotiation changes cost, not tempo.

OUT The scenario

3-period spend path + cost-to-complete, **scenario final cost and VAC**. All arithmetic in C#; the model only narrates.

THE CONTRAST IS THE POINT

planned	realized today	assumed
`BAC ÷ budget qty`	`AC ÷ earned qty`	`the QS's number`

scenario, not a forecast Every figure is a direct arithmetic consequence of the stated assumption — never presented as the validated forecast. For a data-driven number, the copilot routes to the conformal forecaster instead.

UNDER THE HOOD

Production-shaped, not a notebook.

API **ASP.NET Core 8**

Layered Domain / Infrastructure / Core / Web.API — analytics reimplemented in **C#, no Python at runtime**.

UI **React 18 + Vite**

Hand-rolled CSS, **zero UI-framework dependencies**. 12 tabs over one coherent product.

DB **Postgres · RLS**

9 migrations, row-level security, multi-tenant. Each project's data is isolated per user.

AI **Claude agent**

Microsoft Agent Framework over Claude Sonnet 5. Read-only tools, auth resolved before the LLM runs.

~100 xUnit tests — data-contract, leakage, metrics, copilot eval

Every feature passed an **OpenAI Codex review**

WHY IT'S CREDIBLE

Honest where it counts.

📊 Validated, not asserted

Rolling-origin back-test on real history — 45% vs 35% baseline, reported per-fold.

🎯 Honest about limits

Rejected the un-validatable final-cost EAC; labels directional vs validated; single-project scope stated.

🔍 Auditable AI

Tools compute + cite every figure; auth-before-LLM; refuses off-topic. No hallucinated money.

🧩 One coherent product

Watchlist, forecast, variance, stress-test and copilot — a system of record, not four disconnected demos.

THANK YOU

See cost trouble one period early.

A validated early-warning watchlist, a calibrated forecaster, variance attribution, a pre-award stress test, and an auditable copilot that drafts the corrective actions — over one live EVM system of record.

`Next: ML.NET challenger · more projects`